Project Design Phase

Solution Architecture

Date	30 July 2026	Team ID	Individual / Solo Project
Student	Lella Praneeth	**Project Name**	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks		

Solution Architecture

Solution architecture is a complex process that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Architecture Diagram

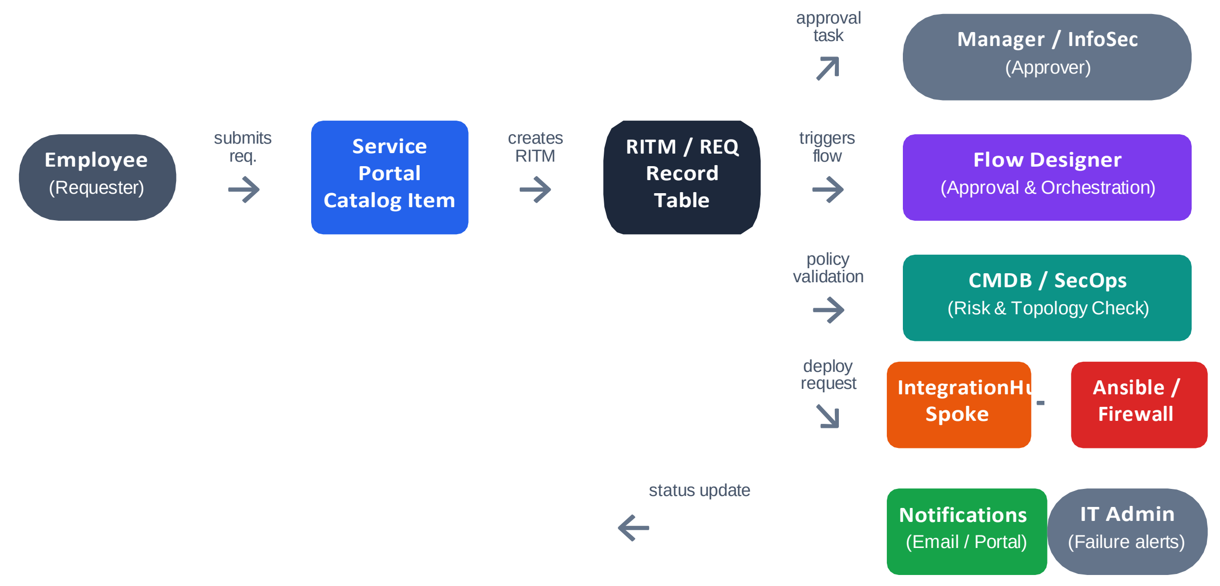

Figure 1: Architecture and data flow of the Automated Network Request Management system

Component Breakdown

Component	Role
Service Portal Catalog Item	Front-end where employees browse and submit network requests, capturing source/destination IP, port, and business justification.
RITM / REQ Table	ServiceNow's native request record structure; stores request details, status, and security audit history end-to-end.
Flow Designer	Orchestrates the entire lifecycle: routes approval tasks to managers/InfoSec, checks CMDB policies, and triggers deployment.

Component	Role
CMDB / SecOps	Validates active network topologies and checks requested ports against baseline security policies before fulfillment.
IntegrationHub Spoke	Bridges ServiceNow with external network controllers to trigger real configurations.
Ansible / Network Controllers	Performs the actual remote configuration on the firewall/router and reports back success/failure.
Notifications	Sends email/portal updates to the employee on approval, rejection, and completion; alerts IT on deployment failures.

Data Flow Summary

An employee submits a network access request through the Service Portal, which creates a RITM record and triggers a Flow Designer flow. The flow routes an approval task to the requester's manager and the InfoSec team. Once approved, the flow queries the CMDB to validate the target configuration, then calls an IntegrationHub spoke that instructs Ansible (or a similar network controller) to remotely apply the firewall or port rule. A confirmation callback updates the RITM, closes the request, and triggers a notification back to the employee, with failure alerts routed to IT if the network deployment does not succeed.

Reference: Adapted from ServiceNow IntegrationHub, Flow Designer, and CMDB documentation patterns.